Jatavath Ajay

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.81
TSEMRS BALANAGAR	Class XII (TELANGANA STATE NOARD OF INTERMEDIATE EDUCATION)	2022	95.40%
ST. MARY'S HIGH SCHOOL	Class X (BOARD OF SECONDARY EDUCATION)	2020	100.00%

INTERNSHIP EXPERIENCE

Intern - AI & Embedded Systems | Amani Automation Pvt. Ltd.

May 2025 - July 2025

- **Developed** a YOLO-based real-time object detection system on **Raspberry Pi 5**, integrating the **Hailo NPU accelerator** to enable efficient AI inference for edge-based computer vision applications.
- **Integrated** AI-based computer vision with **embedded systems and drone hardware**, enabling real-time object detection and processing for **AI-driven UAV applications** in resource-constrained environments.
- **Optimized** AI deployment on **resource-constrained edge devices** through hardware-software integration, improving real-time processing capabilities and supporting efficient execution of AI workloads.

Software Trainee -- VH Mobile Metrics Pvt. Ltd. (360 Technology LLC, USA)

June 2026 - July 2026 | ***AeroInspect AI***

- **Developed** a real-time **YOLOv8-based PPE detection pipeline**, integrating **ByteTrack** for multi-worker tracking and worker-centric safety violation detection during continuous live video monitoring.
- **Engineered** an event-based **safety violation monitoring system** to track violation duration, capture evidence images, and asynchronously transmit detection results to the backend with retry handling.
- **Integrated InsightFace-based worker identification** with confidence-based matching and worker registration, associating detected workers and safety violations with individual profiles during live monitoring.

PROJECTS

RISC-V Assembler | C Programming Project

- **Developed** a RISC-V assembler in **C** to translate assembly instructions into corresponding **hexadecimal machine code**, enabling automated instruction conversion.
- **Implemented** robust **syntax and instruction-format validation**, with error handling modeled on the Ripes Simulator to detect invalid assembly inputs.
- **Designed** a modular architecture with structured **file I/O and debugging components**, improving code maintainability and ease of testing.

Real-Time Object Detection using YOLOv3 | Deep Learning Project

- **Developed** a real-time **object detection pipeline** using YOLOv3 with Python, OpenCV, NumPy, and cv2.dnn for automated object localization across video frames.
- **Implemented** class-wise **Non-Maximum Suppression (NMS)** to eliminate redundant bounding boxes, reducing overlapping detections and refining the final detection results.
- **Processed** video frames and **visualized detection outputs** using FFmpeg and Google Colab video tools, enabling clear analysis of real-time object detection performance.

YOLOv8-SLAM Fusion on Raspberry Pi 5 | Robotics & Embedded AI

- **Deployed** YOLOv8l for **real-time object detection** using the Raspberry Pi Camera on a 64-bit Raspberry Pi platform.
- **Integrated** RTAB-Map for **visual SLAM** with live camera input, enabling spatial awareness in dynamic environments.
- **Fused** object detection and visual SLAM using **ROS 2**, and visualized the combined output in **RViz2** for real-time robotic perception.

Raspberry Pi Camera Control Web Interface | Flask + Picamera2

- **Developed** a real-time **Flask-based camera control interface** using Picamera2 and OpenCV for live video streaming, photo capture, and video recording.
- **Automated** asynchronous photo capture and **H.264-to-MP4 video conversion** using FFmpeg, with integrated file download and deletion functionality.
- **Optimized** camera application performance on **Raspberry Pi 5** by integrating multithreading for non-blocking camera operations and improved responsiveness.

SKILLS

Programming Languages: Python, C, C++, SQL

Machine Learning: Deep Learning, NLP, Computer Vision, Reinforcement Learning

Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face

Tools: Git, Jupyter Notebook, Solid Edge, ArduPilot Mission Planner

Other: Data Structures and Algorithms, RISC-V Assembly

RELEVANT COURSES

Core Courses: Computer Vision, Natural Language Processing, Deep Learning, Reinforcement Learning, Foundations of Machine Learning, Convex Optimization, Programming for AI, Artificial Intelligence, Data Science Analysis, Robotics, Predictive Modelling

Mathematics: Numerical Analysis, Introduction to Bayesian Statistics, Calculus, Differential Equations, Series of Functions, Introduction to Metric Spaces, Matrix Theory, Applied Statistics, Probability and Random Variables

Computer Science: Introduction to Programming, Discrete Mathematics, Data Structures, Computer Architecture, Operating Systems, Database Management Systems, Compilers

Other Courses: Linear Systems and Signal Processing, Digital Fabrication, Marine Robotics

POSITIONS OF RESPONSIBILITY

- **Events Coordinator, *Milan'24***. (The Inter-hostel general championship that encompasses sports, cultural, and technical events.)
- **Publicity Coordinator**, ***Elan & nVision*** (2024-25): Annual techno-cultural fest of IITH.
- **Production and pronites Head**, ***Milan '25.***
- **Events Coordinator**, Inter IIT Sports Meet(***IISM***) 2025.

EXTRACURRICULARS & INTERESTS

- Member of National Service Scheme(NSS)-IITH
- Served as a college people leader from 2021 to 2022.
- ***Certified Remote Drone Pilot*** (Small Rotorcraft - VLOS), Telangana State Aviation Academy (***Govt. of Telangana***, DGCA Approved)